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I. Multiple Choice Questions:

1. A student tunes his guitar by striking a 120 Hertz with a tuning fork, and simultaneously plays the 4th string on his guitar. By keen observation, he hears the amplitude of the combined sound oscillating thrice per second. Which of the following frequencies is the most likely the frequency of the 4th string on his guitar?.

a) 130 b) 117
c) 110 d) 120

2. A transverse wave moves from a medium A to a medium B. In medium A, the velocity of the transverse wave is 500 ms^{-1} and the wavelength is 5 m. The frequency and the wavelength of the wave in medium B when its velocity is 600 ms^{-1}, respectively are

a) 120 Hz and 5 m
b) 100 Hz and 5 m
c) 120 Hz and 6 m
d) 100 Hz and 6 m

3. For a particular tube, among six harmonic frequencies below 1000 Hz, only four harmonic frequencies are given : 300 Hz, 600 Hz, 750 Hz and 900 Hz. What are the two other frequencies missing from this list?.

a) 100 Hz, 150 Hz
b) 150 Hz, 450 Hz
c) 450 Hz, 700 Hz
d) 700 Hz, 800 Hz

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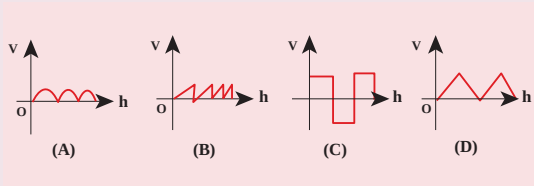
4. Which of the following options is correct?.

A	B
(1) Quality	(A) Intensity
(2) Pitch	(B) Waveform
(3) Loudness	(C) Frequency

Options for (1), (2) and (3), respectively are

a) (B),(C) and (A)
b) (C), (A) and (B)
c) (A), (B) and (C)
d) (B), (A) and (C)

5. Compare the velocities of the wave forms given below, and choose the correct option.



where, v_A, v_B, v_C and v_D are velocities given in (A), (B), (C) and (D), respectively.

a) $v_A > v_B > v_D > v_C$
b) $v_A < v_B < v_D < v_C$
c) $v_A = v_B = v_D = v_C$
d) $v_A > v_B = v_D > v_C$

6. A sound wave whose frequency is 5000 Hz travels in air and then hits the water surface. The ratio of its wavelengths in water and air is

a) 4.30 b) 0.23
c) 5.30 d) 1.23

7. A person standing between two parallel hills fires a gun and hears the first echo